

PERSONAL ENGINEERING PROJECT

MECHATRONICS / ROBOTICS / EMBEDDED CONTROL

PROJECT CASE STUDY

AUTONOMOUS BALL COLLECTION ROBOT

Design, control and prototype development

PERSONAL MECHATRONICS PROJECT

Autonomous ball collection, seesaw traversal and delivery system

CREATED BY

Mevan Shandika Perera

PROJECT FOCUS

Arduino control, actuation and prototyping

COMPLETED

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Executive Summary

Autonomous collection, transport and delivery prototype

This project explores a compact autonomous robot designed to collect three balls of different sizes, transport them securely across a pivoting seesaw, and release them at a destination. The prototype combines a four-wheel drive platform, two servo-actuated retaining arms, and an Arduino Uno running a calibrated time-based control sequence.

CONTROL

Arduino Uno

Sequential motor and servo commands using calibrated timing.

MOBILITY

Four-wheel drive

Four 12 V geared DC motors provide movement and seesaw traversal.

HANDLING

Dual servo arms

Custom grooves and servo arms secure three different ball sizes.

Development outcome

Testing demonstrated successful collection, controlled navigation, and stable seesaw crossing. The table-tennis ball was released successfully; the heavier tennis and racquet balls required more release momentum. The result validated the core mechanical and control architecture while identifying clear opportunities for sensing, voltage regulation, and an assisted release mechanism.

KEY DESIGN DECISION

Separate power supplies were used for the drive motors and servo system to reduce voltage drop and improve repeatability during combined operation.

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System Overview

The prototype uses a deliberately simple architecture: timed Arduino control, differential drive, servo-assisted ball retention, and isolated power for the drive and handling systems.

01**CONTROL**

An Arduino Uno executes one calibrated sequence of forward, reverse, turning, pickup, crossing, and release actions. The loop remains empty after completion.

02**DRIVE**

Four 12 V geared DC motors use differential steering. Reversing wheel polarity enables forward motion, reverse motion, and pivot turns.

03**BALL HANDLING**

Two servo-driven arms retain the balls against dedicated chassis grooves during movement. Servo detachment releases the payload at the destination.

04**POWER**

A 12 V Ni-MH pack supplies the drive motors, while a separate 8 x AA holder powers the servos. Isolation reduces voltage drop during combined operation.

OPERATING SEQUENCE

Approach grid > secure balls > reverse and turn > cross seesaw > align with destination > release payload